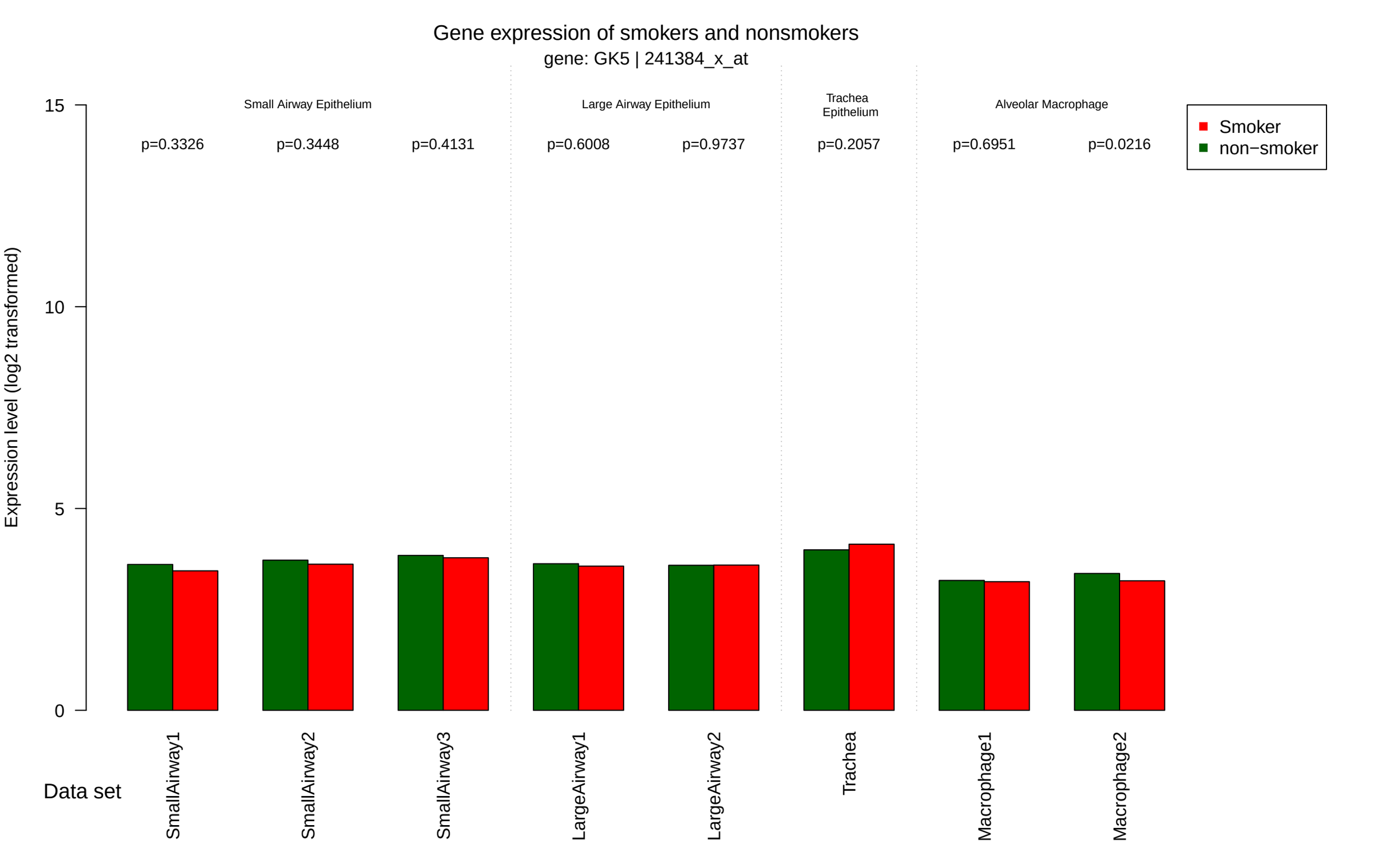
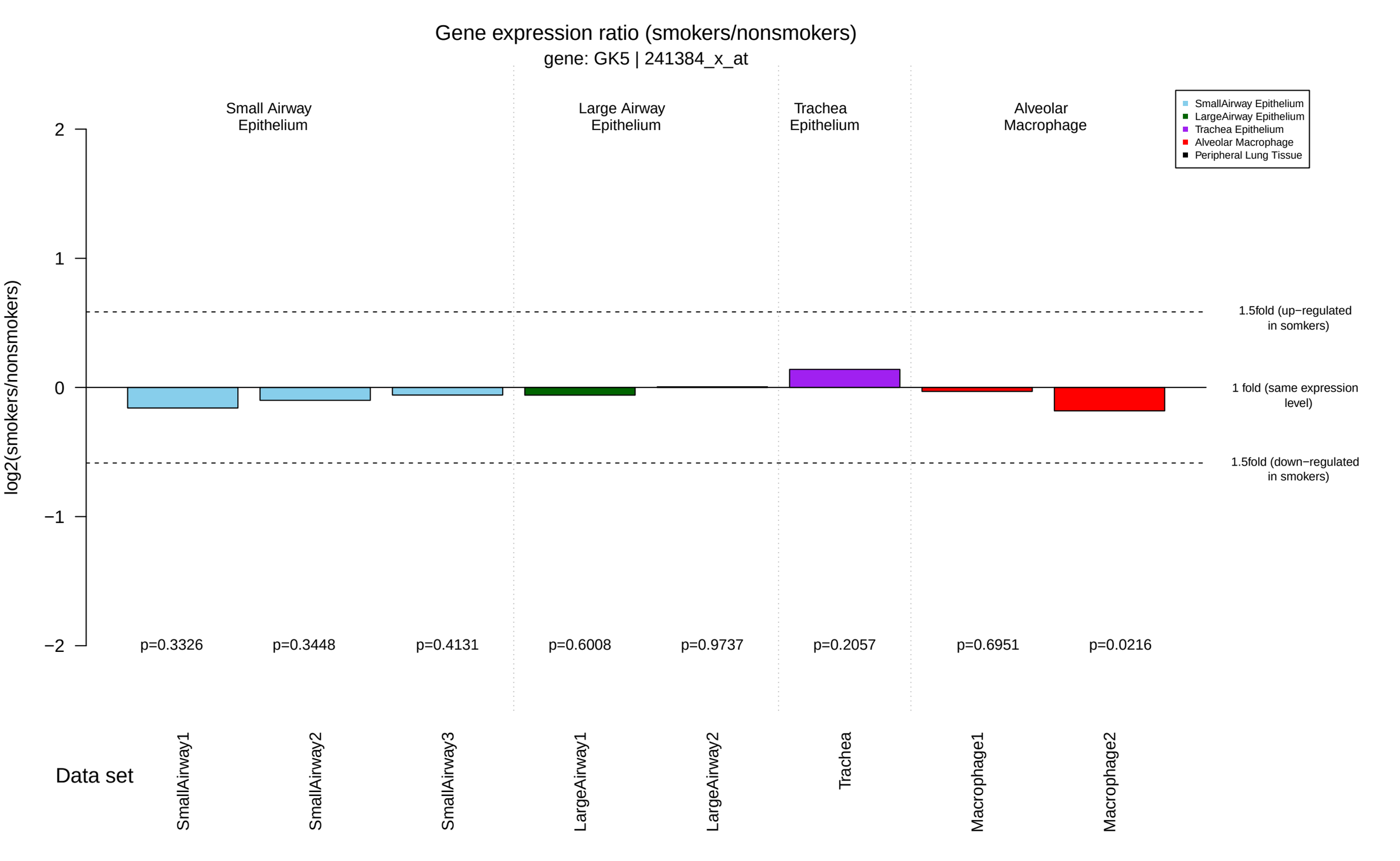
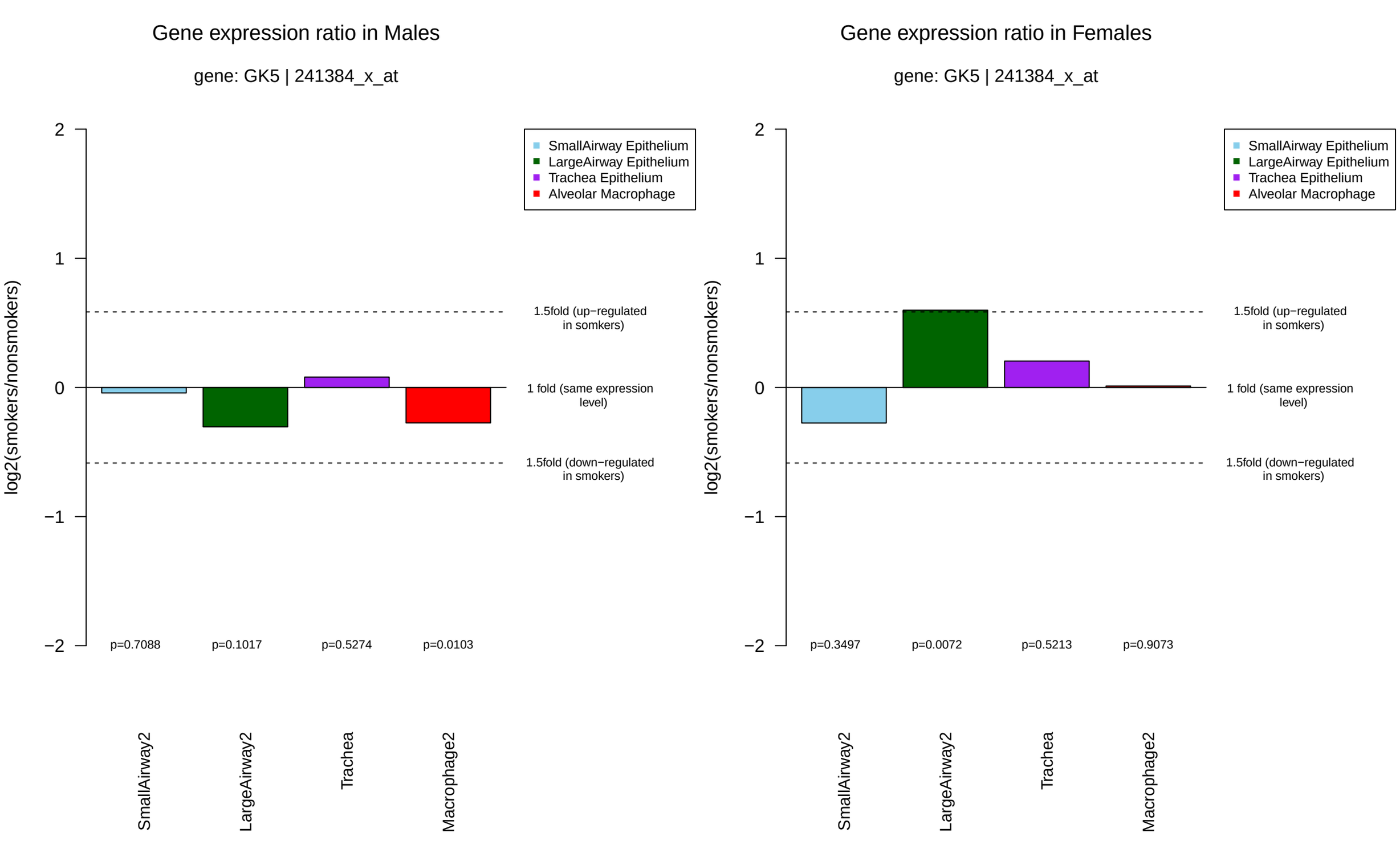
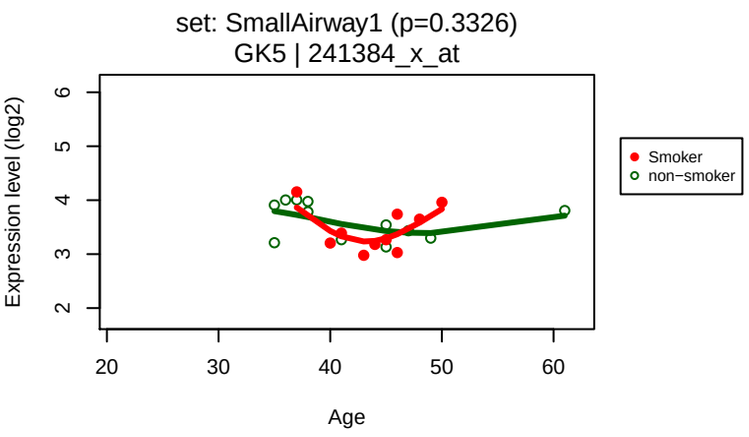
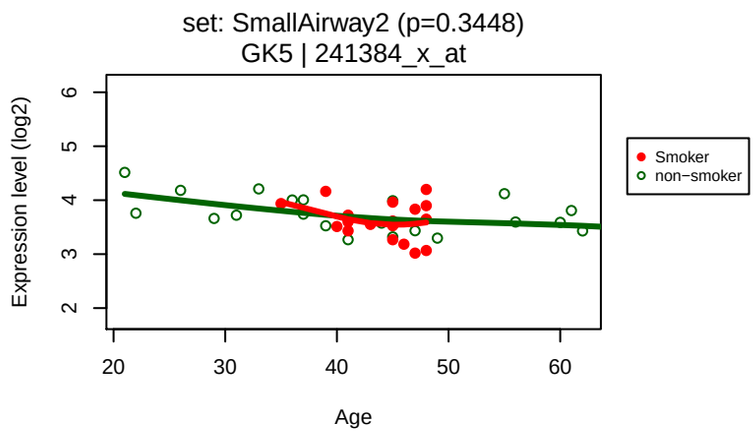
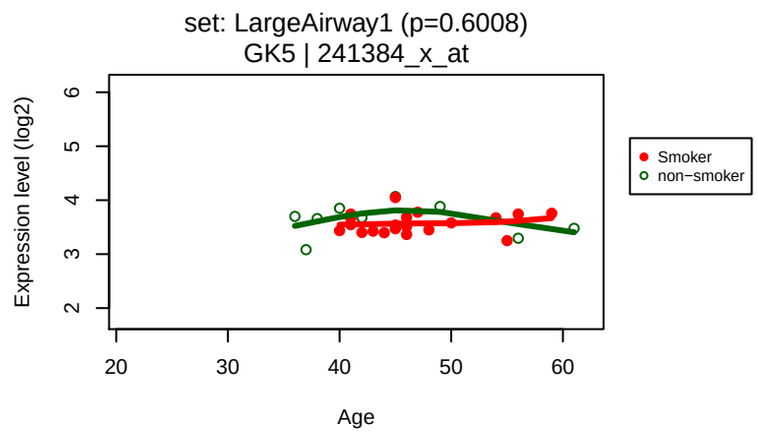

Gene expression of smokers and nonsmokers
gene: GK5 | 241384_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3326
p=0.3448
p=0.4131
p=0.6008
p=0.9737
p=0.2057
p=0.6951
p=0.0216
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

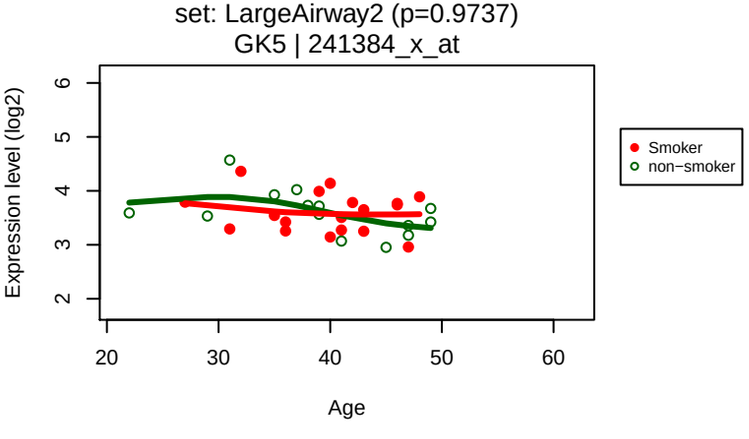

Gene expression ratio (smokers/nonsmokers)
gene: GK5 | 241384_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3326
p=0.3448
p=0.4131
p=0.6008
p=0.9737
p=0.2057
p=0.6951
p=0.0216
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

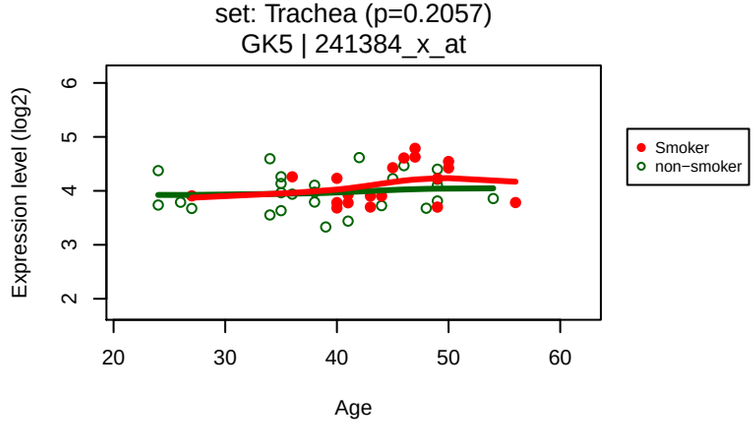

Gene expression ratio in Males
gene: GK5 | 241384_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7088
p=0.1017
p=0.5274
p=0.0103
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GK5 | 241384_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3497
p=0.0072
p=0.5213
p=0.9073
SmallAirway2
LargeAirway2
Trachea
Macrophage2

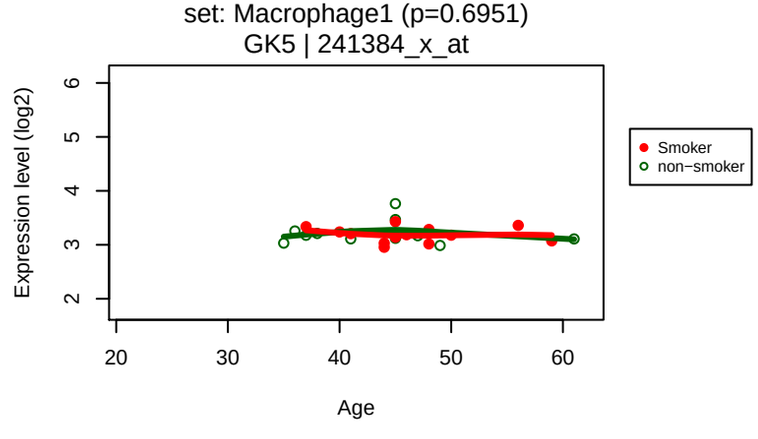

set: SmallAirway1 (p=0.3326)
GK5 | 241384_x_at
Expression level (log2)
Age
Smoker
non−smoker

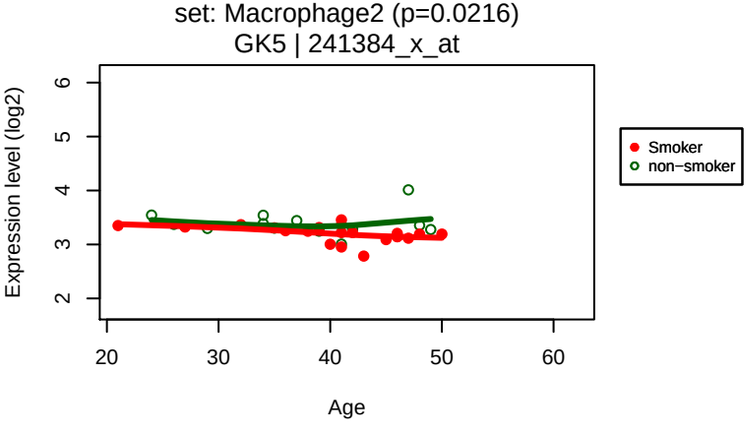

set: SmallAirway2 (p=0.3448)
GK5 | 241384_x_at
Expression level (log2)
Age
Smoker
non−smoker

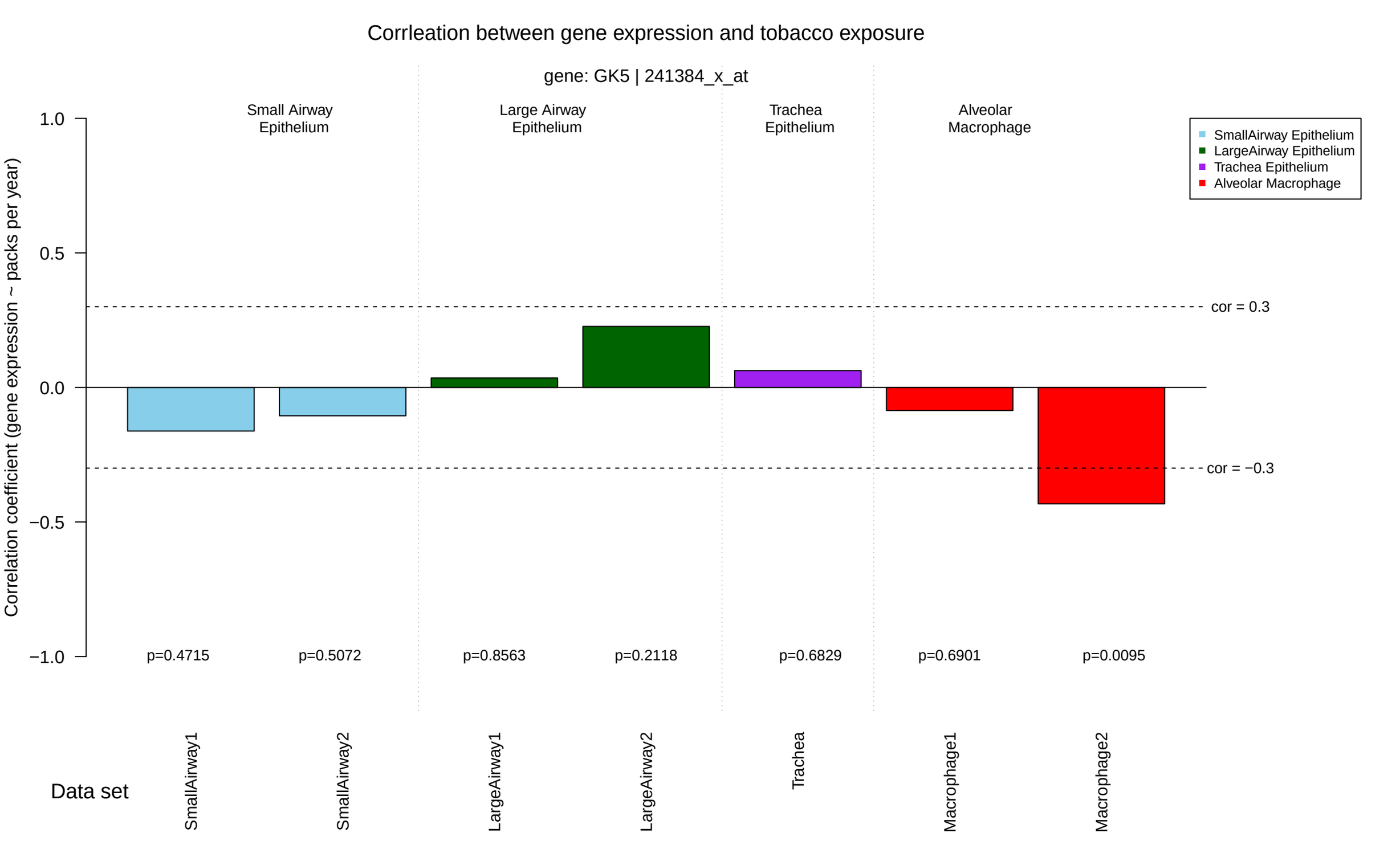

set: LargeAirway1 (p=0.6008)
GK5 | 241384_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9737)
GK5 | 241384_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2057)
GK5 | 241384_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6951)
GK5 | 241384_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0216)
GK5 | 241384_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GK5 | 241384_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4715
p=0.5072
p=0.8563
p=0.2118
p=0.6829
p=0.6901
p=0.0095
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2